

Error (Uncertainty) Analysis

HW IS PAGES 6/7 EXERCISES

I. Reporting Uncertainty

The uncertainty in a reported result or measurement is a statement about how well we know the value of our measured (calculated or determined) quantity. The uncertainty may be called the error or even standard deviation in the value. This supersedes anything you thought you knew about significant figures.

For example we may report the speed of light as:

$$c=(2.994\pm0.032) \times 10^8 \text{ m/s}$$

The number 0.032 is the uncertainty in my measurement or determination (some people say estimate of uncertainty) the speed of light. In general an uncertainty is kept to 1 or 2 significant figures (2 if the first non-zero digit is a 1,2,3 and 1 if otherwise).

This is a result placed in the abstract to my paper as “my measured result”. I also indicate that my determination is about 0.1 error bars (uncertainties, or standard deviations) away from the accepted value of $2.997 \times 10^8 \text{ m/s}$ in air.

The number 0.032 has to do with the PRECISION of my experiment. Whereas the 0.003 off from the “accepted result” has to do with the ACCURACY of the measurement. It is not possible to discuss accuracy when one is measuring an unknown quantity for which there is no standard accepted or defined value. Precision tells us how well we can reproduce our measurement, where accuracy tells us how close to the standard we are. A topic relating to instrument accuracy is “calibration” where results are checked against known standards.

The number 0.032 should relate somehow to basic measurements that went into determining our value for the speed of light “c”. For example we may be using displacement and elapsed time to yield the speed of light. Each, individual displacement and time, have their own uncertainties which I will typically estimate (not guess). Estimates are based on your experience with instruments, techniques, similar experiences---and they are not “gusses”.

This document will help understand how to determine final uncertainties in measured results and how that uncertainty relates to the basic quantities you measure directly. Some of the discussion is from documents you may have seen in introductory physics. The work you will do is in section IV, “further discussion” which adds some more advanced results (and you may have seen some of this in nuclear labs).

II. Types of Experimental Error

No physical measurements can be made to infinitely high precision; therefore an estimate of error must be included with *each* measurement. In the collection of data, two types of experimental errors, systematic and random usually contribute to the error in the measured quantity.

Systematic errors are produced by a well-defined cause which affects every measurement in exactly the same way. Errors of this type cause measured values to be consistently too high or consistently

too low. Systematic errors may be eliminated once this well defined cause is isolated. Systematic errors may be of four kinds:

Let's consider some examples. Take a frayed piece of string and measure the length. There is uncertainty as to where you will call the left edge and the right edge. After different classmates measure the length a thousand times you will get a distribution of measurements. If these measurements (of the same identical quantity, using the same instruments and method) are randomly scattered about a single value, then a histogram of number of measurements vs. length will typically map out a pretty good bell curve, or Gaussian distribution of values. It is not a perfect bell curve, but will get better if we make 100,000 measurements rather than 1000. We can, to a very good approximation identify the peak of the curve (this is our best value for the length) and a half width for the curve (the standard deviation, uncertainty, or error). We often make a well educated estimate of such an uncertainty since we are often familiar with the instrument (eg. a ruler or meterstick).

If using the same frayed piece of string as above, we were given a meter stick whose numbers were painted in incorrect positions by the manufacturer, then our determination of length would be incorrect. This is an example of a systematic error. Other examples may be a stopwatch that runs too fast or slow, or a scale that is not calibrated. Systematic errors can often be calibration errors.

We may assign the following labels to various types of uncertainties:

1. *Instrumental*. For example, a poorly calibrated instrument such as a thermometer that reads 101°C in boiling water and 1°C in ice water would give measured temperatures that are consistently too high.
2. *Observational*. For example, parallax in reading a meter scale.
3. *Environmental*. For example, an electrical power "brown out" that causes measured currents to be consistently too low.
4. *Theoretical*. Due to simplifications of the model system or approximations in the equations describing it. An example of this could be ignoring frictional forces in one's calculations even though a frictional force acts during the experiment. The experimental and theoretical results will consistently disagree.

In principle, an experimenter wants to *identify and eliminate systematic errors*.

But even without systematic errors, the values obtained in a physical measurement will always lie within a range of values, rather than yield a unique value. These irregular deviations of a measurement are said to be random errors. Random errors are positive and negative fluctuations that cause about one-half of the measurements to be too high and about one-half of the measurements to be too low. Sources of random errors cannot always be identified.

Random errors, unlike systematic errors, can often be quantified by statistical analysis; therefore, the effects of random errors on the quantity or physical law under investigation can often be determined.

The distinction between random errors and systematic errors can be illustrated with the following example. Suppose the measurement of a physical quantity is repeated five times under the same conditions. If there are only random errors, then the five measured values will be spread about the "true value"; some will be too high and some will be too low as shown in Figure 1. If there are also systematic errors, then the five measured values will be spread, not about the true value, but about

some displaced value as shown in Figure 2. Note that the *accuracy* of the experiment is a measure of the systematic error in the experiments and that the *precision* of the experiment is a measure of the random error in the experiment.

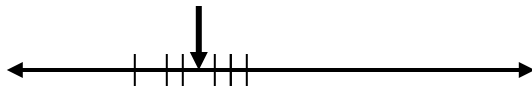


Figure 1 (Only random errors)

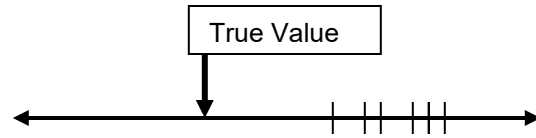


Figure 2 (Random and systematic errors)

III. Absolute and Percentage Errors – An Approximation to Error Analysis

Approximating errors and their analysis is often based on maximum pessimism. Consider the examples below:

Absolute Error Examples	$397 \pm 2.0\text{cm}$	$32 \pm 2.0\text{cm}$	$1.03 \pm 0.010\text{s}$	$44.89 \pm 0.010\text{s}$
Corresponding Percent Errors:	$397\text{cm} \pm 0.5\%$	$32\text{cm} \pm 6\%$	$1.03\text{s} \pm 1.0\%$	$44.89\text{s} \pm 0.020\%$

In each case, the absolute error defines the maximum amount the value could be different from the true value. It is a *guarantee made by the experimenter* that the true value is between the limits implied by the quoted error or uncertainty. In the first example, the experimenter claims that the true value is *definitely* between 395cm and 399cm.

We often have to measure several quantities in order to determine the desired result. For example, to measure the area of my back yard, I have to measure both length and width. And both length and width each have error. So the question is – how does the error in the length and width impact the error of the product (the area)? We call this error propagation – in this section you will develop “rules” for error propagation. Pay close attention since you will be using these rules throughout the course.

PROPAGATING ERROR – ADDITION AND SUBTRACTION

To learn how to propagate error when **adding**, work the following example:

EXAMPLE 1: A student walks $2.0 \pm 0.1\text{m}$, stops and then walks another $3.5 \pm 0.2\text{m}$ in the same direction.

a. What is the *smallest* distance the student could possibly be from the starting point? (Minimum possible total)

b. What is the *largest* distance the student could possibly be from the starting point (Max total)?

c. Use these answer to develop a “rule” for error propagation when adding numbers. (You should be able to show that the rule for subtraction is the same!) USE MIN AND MAX TO EXPRESS ANSWER—TOTAL=NUMBER + OR – ERROR.

d. Write your “rule” here:----WHAT TO DO WITH INDIVIDUAL ABSOLUTE ERRORS TO GET SAME RESULT AS C.

PROPAGATING ERROR – MULTIPLICATION AND DIVISION

To learn how to propagate error when multiplying, work the following example:

EXAMPLE 2: You are trying to determine the area of the floor of a rectangular closet since you will be replacing the carpet. You measure the length to be $2.0 \pm 0.1\text{m}$ and the width to be $4.0 \pm 0.1\text{m}$.

a. What is the *maximum* possible area based on the measurements and associated errors?

b. What is the *minimum* possible area based on the measurements and associated errors?

c. What is the best estimate of the area – and what is your error estimate for the area?
AREA=NUMBER + OR – ERROR NUMBER. (ERROR FOUND USING MAX, MIN)

d. What is the percent error of the area? (USING ERROR NUMBER FROM C)

e. What is the percent error in the depth measurement?

f. What is the percent error in the width measurement?

g. Now, comparing your answers from parts *d*, *e* and *f*, can you come up with a rule for error propagation when multiplying (the rule for division by the way is the same!). THE PERCENT ERROR IN AREA RELATES HOW TO INDIVIDUAL PERCENT ERRORS.

IV. Further discussion

The discussions above give you a sense for how to handle experimental uncertainties. Greater formalism leads to a more general statement of uncertainty in a measured quantity. When working with some measured result such as a function $f(x,y,z)$ we can denote the uncertainty as $\delta f(x_1,y_1,z_1)$. Where x_1,y_1,z_1 are any variables (maybe position, but maybe volume, pressure, and temperature). The uncertainty may be thought of as the range of results you would see if you measured $f(x_1,y_1,z_1)$ repeatedly. Other jargon for uncertainty is “error”, or “standard deviation”.

In multivariable calculus we may say that the variation in a value given by a function is $df(x,y,z)$. If we can determine an expression for df , we can evaluate at x_1,y_1,z_1 . The symbol df ---evaluated numerically---is the same as δf .

Mathematically

$$df(x,y,z) = \left(\frac{\partial f}{\partial x}\right) dx + \left(\frac{\partial f}{\partial y}\right) dy + \left(\frac{\partial f}{\partial z}\right) dz \quad 1$$

Each of the terms in brackets is called a partial derivative. It means you take the derivative of the function $f(x,y,z)$ with respect to the variable such as x only. This gives the rate of change of the function with respect to x . The dx refers then to an experimental uncertainty in x , and the uncertainty in f is given by the left hand side (after evaluating all terms at some point x_1,y_1,z_1).

In general we modify the above equation to accommodate the fact that uncertainties will usually gang up in different directions (uncorrelated). **The result is the “holy grail” formula for error analysis and adds each of the terms in the above equation in “quadrature”.**

$$df(x,y,z) = \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 dx^2 + \left(\frac{\partial f}{\partial y}\right)^2 dy^2 + \left(\frac{\partial f}{\partial z}\right)^2 dz^2} \quad 2$$

Recall that you will need to know the function f , evaluate it at particular x,y,z , after taking derivatives,

Some comments: In order to use this method for determining uncertainty, you must have a model or theoretical expression describing the behavior of the quantity $f(x,y,z)$. However the behavior can be determined experimentally. Each term in our formula can be measured, though this may be tedious. If you can graph $f(x,y,z)$ vs. x , then you know the partial derivative of f with respect to x (a slope from the graph at that point) and can use that to determine uncertainty.

There are many many other nuances to working with uncertainties that will be explored at a later time. Right now use the above equation when possible to estimate uncertainties in quantities “ f ”.

Exercise 1. Show using equation 2 that if $f(x,y,z) = x^2y/z$ where $dx = \sigma_x$, $dy = \sigma_y$, $dz = \sigma_z$ then the uncertainty of f , σ_{total} becomes. (This is simply adding fractional uncertainties that we had as a rule for multiplication). n_x, n_y, n_z , respectively here are the powers to which x , or y , or z is raised in the simple polynomial expression (eg. $n_x = 2$, and the others are 1).

$$\frac{\sigma_f}{f(x,y,z)} = \sqrt{\left[\frac{n_x \sigma_x}{x}\right]^2 + \left[\frac{n_y \sigma_y}{y}\right]^2 + \left[\frac{n_z \sigma_z}{z}\right]^2} \quad 3$$

Exercise 2. Find the expressions for uncertainty in the measured intensity, σ_I of a beam of photons incident on a material. We know that $I = I_0 e^{-\mu x}$ describes the attenuation of a photon beam in a material with attenuation coefficient μ . The measured quantities with their uncertainties are $\mu \pm \sigma_\mu$, $x \pm \sigma_x$ and $I_0 \pm \sigma_{I_0}$. Does your result depend on x , I_0 , μ ? You are finding an expression for σ_I .

Exercise 3.

Consider the following data for position and time. We will calculate a velocity from the measured values of position and time for a “pulse” of light.

$t(\text{s}) \pm 0.1 \text{ ns}$	$x(\text{m}) \pm 0.005 \text{ m}$
2.4E-10	0.000
8.8E-10	0.200
1.54E-9	0.400
2.26E-9	0.600
2.96E-9	0.800
3.56E-9	1.000
4.2E-9	1.200
4.92E-9	1.400
5.7E-9	1.600
6.32E-9	1.800
6.76E-9	2.000

We can obtain estimates of the uncertainty in the velocity in several ways.

- a) First make a plot of x vs. t and use the maximum and minimum slopes from data at the endpoints only. Your uncertainty should be $|\text{max slope} - \text{min slope}|/2$. Your reported result should look like. You should determine a best number for c and explain how.

$c = \text{_____} \pm \text{uncertainty}$

- b) Statistical analysis of data sets indicates that more measurements should reduce the uncertainty of our velocity determination. In part a we used only 2 data points to determine the uncertainty but we have 11 data points. In counting experiments the uncertainty in a counting value is proportional to $1/\sqrt{N-1}$ and typically N is sufficiently large that $N-1$ gets treated as N . So, as a rough approximation to the uncertainty in the speed of light with the given data, take your result from part a and reduce it by the $\sqrt{11}$ and report your new result.

$c = \text{_____} \pm \text{uncertainty part b}$

remember this is an approximate method.

- c) Another method you will use frequently is to use a linear regression, (trend line, trend analysis, linear fit, least squares fit,) to determine a slope (speed here) and uncertainty in the slope. The uncertainty in most such fitting routines assumes equal weighting of uncertainty from all data points. The uncertainty comes from the statistical fluctuations of your data around a perfect line. The data does not fall perfectly on a straight line. Use a fitting routine (your choice---origin, excel with linest installed, matlab toolbox, mathematica, etc)---ORIGIN is Colbert's PREFERRED. Fit the data set and report results. Use what software you like—but it must give an uncertainty in the resulting parameters of fit (here slope and intercept)

$c = \text{_____} \pm \text{uncertainty part c}$

How does the value from fitting (with no knowledge of your experimental uncertainties), which only considered statistical fluctuations in data, compare to uncertainty results from part b?

- d) In real research you would probably take the time to insert additional information into your fitting program before doing a fit. Along with x and t , you would have a column for uncertainty in x , and a column for uncertainty in t . In general these may not all be equally weighted. I will not have you do this.

If you have equal weighting of uncertainties for your data (usually this is percent or fractional) then there is no need to enter uncertainties into a fitting program. The result is the same for any equal weighting. If you use the same absolute uncertainty, such as ± 1.0 cm for each data point, the weighting is approximately equal if your data does not range over many orders of magnitude---and one may use a typical approximation of equal weighting.